

DTU



Model-based Machine Learning

Temporal models

Case study: Modeling freeway occupancy rates in San Francisco

Autoregressive models

State-space models

Multivariate state-space models

Road to MBML: where are we?

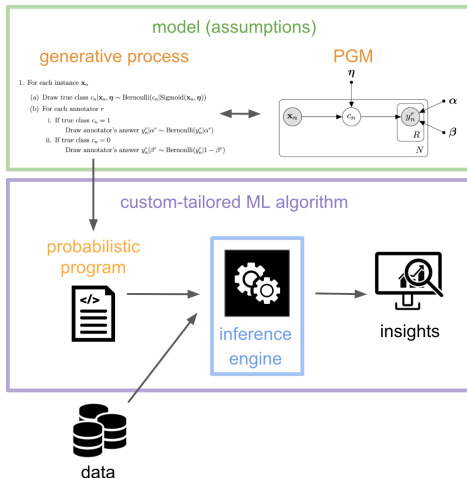
Foundations of PGMs and probabilistic programming

Probabilistic modelling in various contexts - your modelling toolbox

Bayesian inference (exact and approximate)

Week	Topic
1	Intro to the course + Prob. review
2	PGM foundations
3	PGM foundations II
4	Freq. vs Bayesian + Prob. Prog. + Mixture models
5	Regression models
6	Classification and Hierarchical models
7	Temporal models
8	Topic Models
9	Markov-chain Monte Carlo (MCMC)
10	Variational inference
11	Generative models
12	Gaussian processes
13	Project support

Model-based Machine Learning



At the end of this lecture, you should be able to:

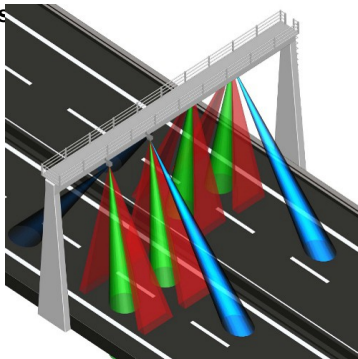
- Explain what Markov models are and their underlying assumptions
- Explain the underlying concepts and assumptions behind state-space models
- Explain the difference between forecasting and imputation
- Relate different ways of modelling the temporal dependency between variables and justify their suitability for a problem
- Implement different temporal modelling techniques in Pyro

Modeling freeway occupancy rates in San Francisco

- Sensors measure occupancy rate (between 0 and 1) of different car lanes
- Time-series of measurements every 10 minutes (144 observations per day)
- Multivariate time-series data: one time-series per sensor

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- Time-series of measurements every 10 minutes (144 observations per day)
- Multivariate time-series data: one time-series per sensor
- Goal: **model freeway occupancy rates**
- Some possible applications:
 - Predict future occupancy rates for better routing
 - Identify problems and send alerts to road users
 - Understand drivers' behaviours



Modeling freeway occupancy rates in San Francisco (cont'

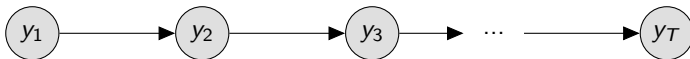
- Let's start thinking about the graphical model...
- We shall start by considering **only one sensor**
- Let y_t denote the occupancy rate in that sensor at time t



- How should we connect these (dependencies)?

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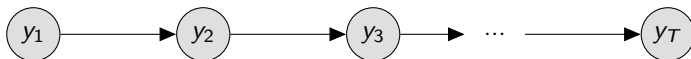
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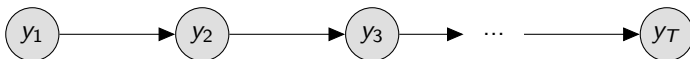
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- What other variables could we have considered?
 - Seasonal information
 - Day of the week
 - Weather data
 - Information from other sensors
 - ...but for now we shall consider only data from the time-series $\{y_1, \dots, y_T\}$

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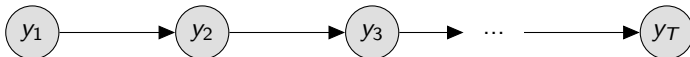
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 - ...but for now we shall consider only data from the time-series $\{y_1, \dots, y_T\}$
- How should we model the dependency of y_t on y_{t-1} ?

Markov models (or Markov chains)

- Simplest mathematical models for random phenomena evolving in time
- Assume that **the present depends only on the recent past**

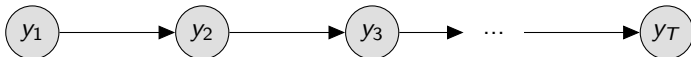
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- Joint distribution of all variables factorizes as

$$p(y_1, y_2, \dots, y_T) = p(y_1) \prod_{t=2}^T p(y_t | y_{t-1})$$

- This only requires us to specify a **transition probability** $p(y_t | y_{t-1})$ (well, and some prior distribution for the first observation y_1)

Note

Transition probabilities are time-independent (same transition distribution for all t)

Autoregressive (AR) models

- Standard and widely used Markov model of continuous observations
- Assume that each y_t is a **linear function** of the M previous observations (plus some Gaussian-distributed observation noise term $\epsilon \sim \mathcal{N}(0, \sigma^2)$)

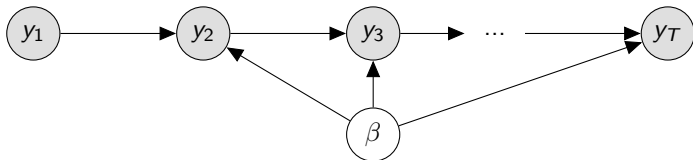
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- Graphical model for (Bayesian) autoregressive model of order 1 (AR1)

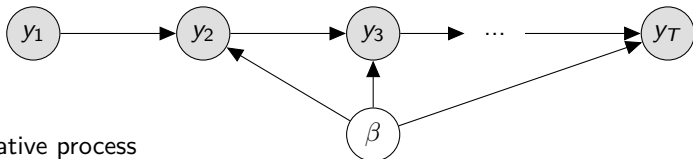


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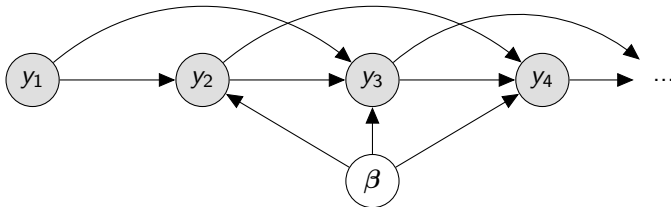
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- Generative process
 - 1 Draw transition coefficient $\beta \sim \mathcal{N}(\beta|0, \lambda)$
 - 2 Draw first observation $y_1 \sim \mathcal{N}(y_1|\mu_0, \sigma_0^2)$
 - 3 For each time $t \in \{2, \dots, T\}$
 - a Draw observation $y_t \sim \mathcal{N}(y_t|\beta y_{t-1}, \sigma^2)$

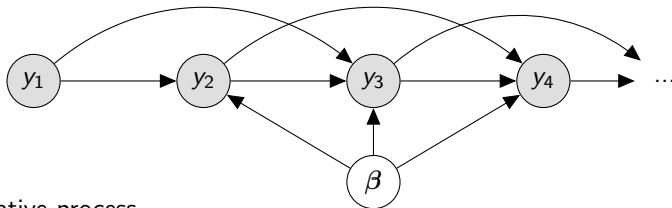
Autoregressive (AR) models: higher-level dependencies

- Graphical model for (Bayesian) autoregressive model of order 2 (AR2)



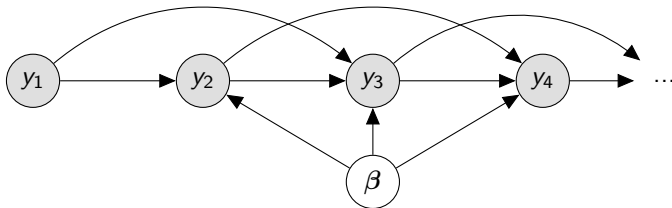
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 - 3 Draw second observation $y_2 \sim \mathcal{N}(y_2 | \beta_1 y_{t-1}, \sigma^2)$
 - 4 For each time $t \in \{3, \dots, T\}$
 - a Draw observation $y_t \sim \mathcal{N}(y_t | \beta_1 y_{t-1} + \beta_2 y_{t-2}, \sigma^2)$

Autoregressive (AR) models: summary so far...



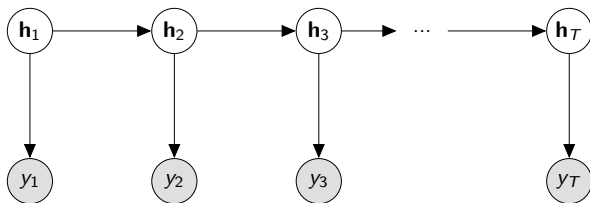
- In autoregressive models, each observation y_t is a linear function of the M previous observations $\{y_{t-1}, \dots, y_{t-M}\}$ plus some noise
- But, if $\{y_{t-1}, \dots, y_{t-M}\}$ are themselves noisy observations (e.g. due to measurement noise in sensors), aren't we building up noise over time?
- Perhaps there is a better way to model temporal data...

State-space models (SSMs)

- In AR models, only the observations $\{y_1, \dots, y_T\}$ are modeled explicitly
- More general models of time-series introduce an extra set of unobserved (hidden) variables $\{\mathbf{h}_1, \dots, \mathbf{h}_T\}$ from which the observations are assumed to be generated

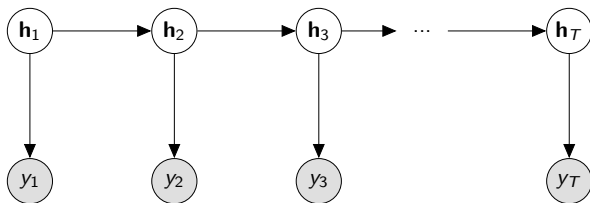
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- Notice that the latent **states** $\mathbf{h}_t$ can be vector-valued!
- Many physical processes can be expressed in this state-space framework

Linear dynamical systems (LDSs)

- State-space models require us to specify two probability distributions:
 - Transition probability: $p(\mathbf{h}_t | \mathbf{h}_{t-1})$
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- **B**, **R**, **c** and σ are often treated as parameters rather random variables
 - Maximum likelihood estimation (MLE) or maximum-a-posteriori (MAP) rather than a fully Bayesian approach
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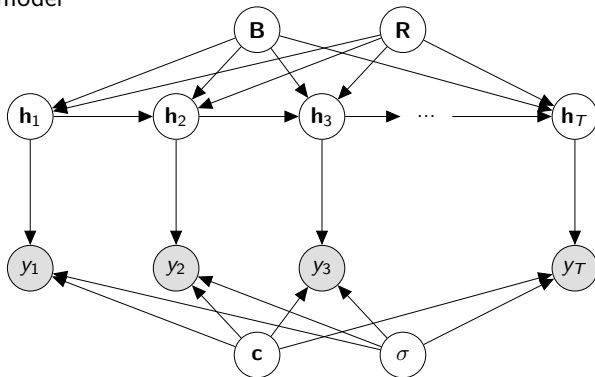
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- We shall consider a fully Bayesian approach
 - Assign priors to $\mathbf{B}$, $\mathbf{R}$, $\mathbf{c}$ and σ
 - Use Bayesian inference to compute respective posteriors

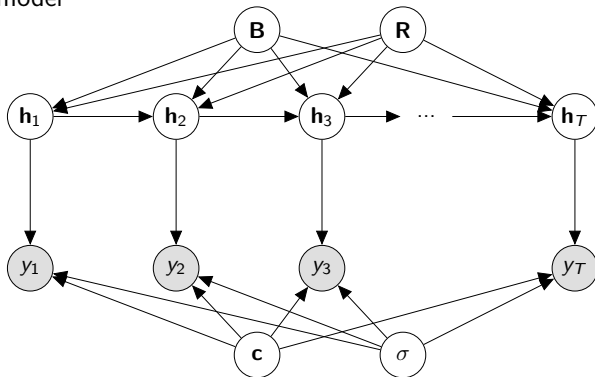
Linear dynamical systems (LDSs)

- Graphical model



Linear dynamical systems (LDSs)

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- The goal of inference is to compute the posterior distribution over all the latent variables in the model: $\{h_1, \dots, h_T\}$, B , R , c and σ
 - Can be very challenging (e.g. identifiability issues)!
 - We often need to impose some restrictions on the structure of B , R , c .

AR models as linear dynamical systems (LDSs)

- By appropriately defining the latent states, many popular time-series models can be cast as an LDS
- E.g. an autoregressive model of order k (AR- k) can be written as:

$$\underbrace{\begin{pmatrix} h_t \\ h_{t-1} \\ \vdots \\ h_{t-k+1} \end{pmatrix}}_{\mathbf{h}_t} = \underbrace{\begin{pmatrix} \beta_1 & \beta_2 & \cdots & \beta_k \\ 1 & 0 & 0 & 0 \\ 0 & \ddots & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}}_{\mathbf{B}} \underbrace{\begin{pmatrix} h_{t-1} \\ h_{t-2} \\ \vdots \\ h_{t-k} \end{pmatrix}}_{\mathbf{h}_{t-1}} + \begin{pmatrix} \tau_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

with the emission probability defined as

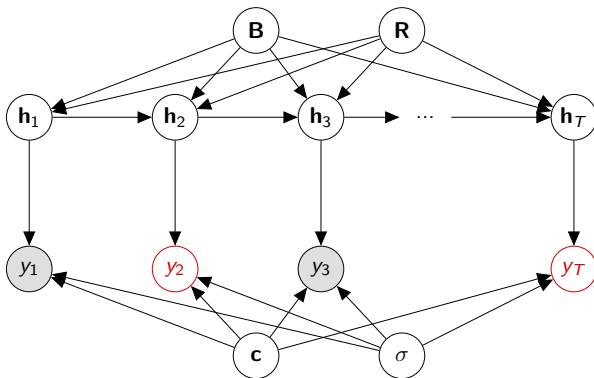
$$y_t = \underbrace{(1 \quad 0 \quad \cdots \quad 0)}_{\mathbf{c}} \mathbf{h}_t + \epsilon_t$$

where $\tau_t \sim \mathcal{N}(0, r)$ and $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$

- Notice the restrictions that we imposed on the structures of $\mathbf{B}$ and $\mathbf{c}$

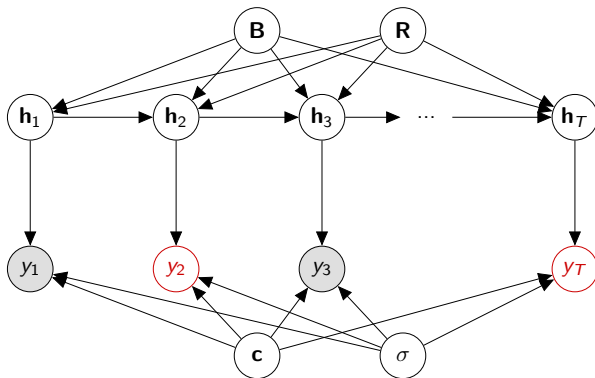
Missing data

- What if we have **missing** observations?



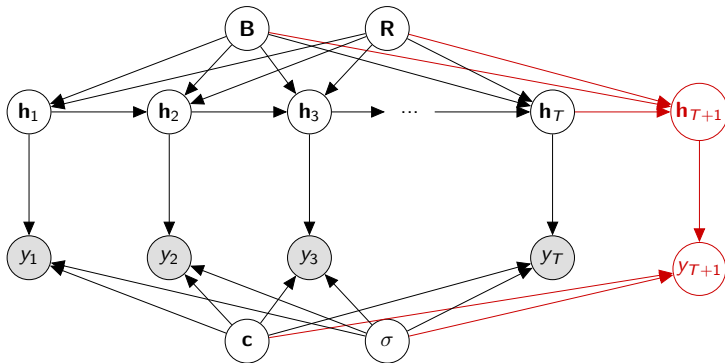
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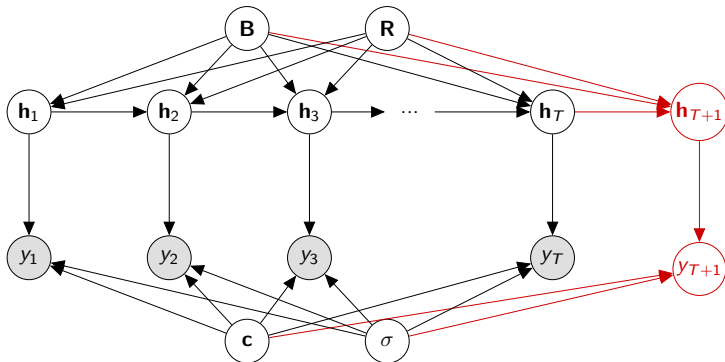


- Just treat the **missing observations** as latent variables
 - Marginalize out their values
 - Compute their posterior distribution

- What if we want to predict the value of y_{T+1} ?



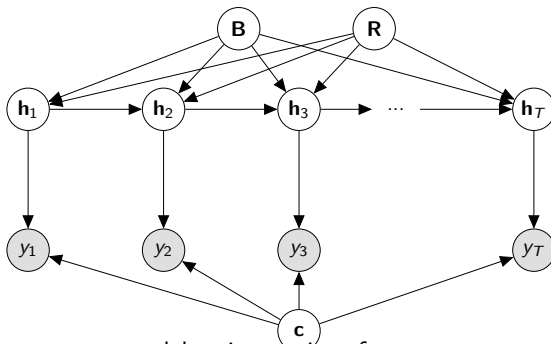
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- Extend** the model for one (or more) time-steps with latent variables
 - Compute the posterior distribution over y_{T+1}

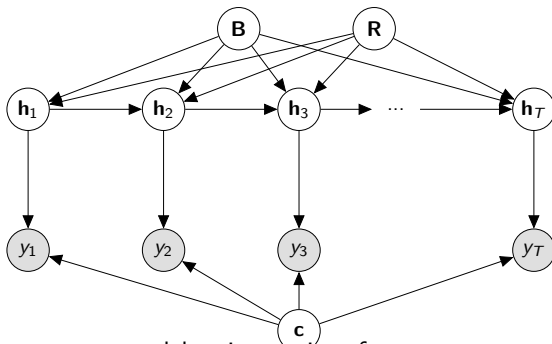
- Ancestral sampling from Bayesian LDS model
 - See "06 - Temporal models - Part 1.ipynb" notebook
 - Expected duration: 10 minutes
- Bayesian LDS model of freeway occupancy rates: forecasting
 - See "06 - Temporal models - Part 2.ipynb" notebook
 - Expected duration: 45 minutes

Extensions: Poisson LDSs



- Suppose we want to model a time-series of counts
 - E.g. forecasting the number of arrivals/departures at a metro station

Extensions: Poisson LDSs

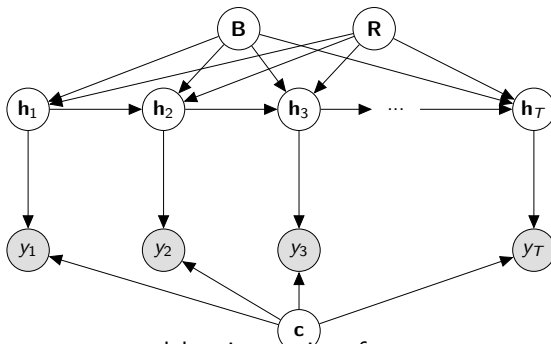


- Suppose we want to model a time-series of counts
 - E.g. forecasting the number of arrivals/departures at a metro station
- A **Poisson distribution** could be a better choice for the emission probabilities

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$$y_t \sim \text{Poisson}(y_t | \exp(\mathbf{c}^\top \mathbf{h}_t))$$

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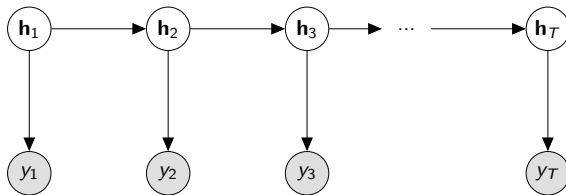
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- In fact, as with generalized linear models (GLMs), we could use **any** general

Extensions: Linear dynamical systems (LDSs) with inputs

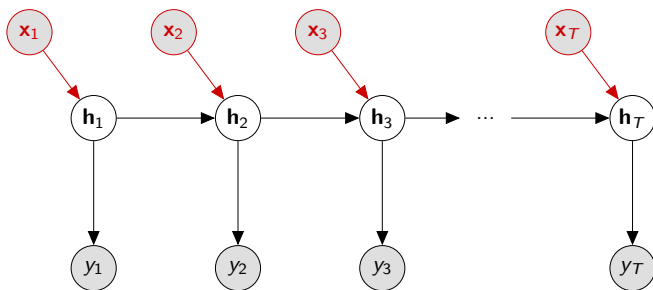
- Graphical model representation of a linear dynamical system:



(for simplicity, we omitted the variables $\mathbf{B}$, $\mathbf{R}$, $\mathbf{c}$ and σ from the graph)

- What if we have additional input information available at each time t that can help explain the observations y_t ?
 - In our case study of freeway occupancy rates, this could be event information, weather, seasonal information, etc.

Extensions: Linear dynamical systems (LDSs) with inputs



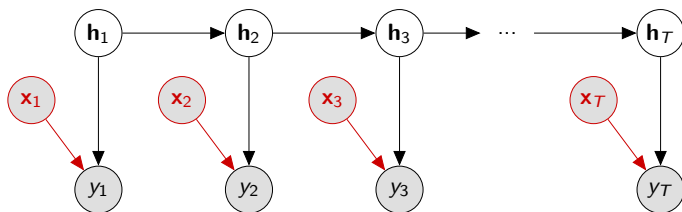
(for simplicity, we omitted the variables $\mathbf{B}$, $\mathbf{R}$, $\mathbf{c}$, σ and $\mathbf{W}$ from the graph)

- x_t are **external inputs** at time t
- Transition probability becomes:

$$h_t \sim \mathcal{N}(h_t | \mathbf{B}h_{t-1} + \mathbf{W}x_t, \mathbf{R})$$

Extensions: Linear dynamical systems (LDSs) with inputs

Or, alternatively:



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- $\mathbf{x}_t$ are **external inputs** at time t
- Emission probability becomes:

$$y_t \sim \mathcal{N}(y_t | \mathbf{c}^T \mathbf{h}_t + \mathbf{W} \mathbf{x}_t, \sigma^2)$$

- What are the implications? How do they differ from the previous slide?

Extensions: Extended Kalman filter (EKF)

- So far we have assumed linear relations between states and from state to observation

$$\mathbf{h}_t \sim \mathcal{N}(\mathbf{h}_t | \mathbf{B}\mathbf{h}_{t-1} + \mathbf{W}\mathbf{x}_t, \mathbf{R})$$

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- The *extended Kalman filter* (EKF) relaxes this assumption by allowing these relations to be defined by arbitrary **differentiable functions** f and g

$$\mathbf{h}_t \sim \mathcal{N}(\mathbf{h}_t | f(\mathbf{h}_{t-1}, \mathbf{x}_t), \mathbf{R})$$

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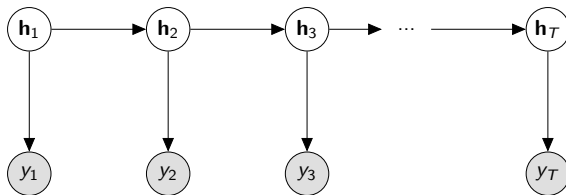
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- f and g can even be neural networks! (search for “Deep K”)
- EKF allows for a **non-linear** version of the Kalman filter
- Extremely popular in navigation systems and GPS (e.g. your phone uses it all the time!)
- Specialized algorithms have been developed for learning and performing inference in EKFs - not the focus of this course! We will keep relying on Pyro

Extensions: Different regimes

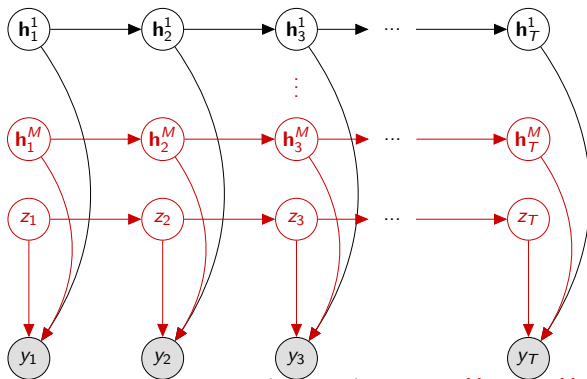
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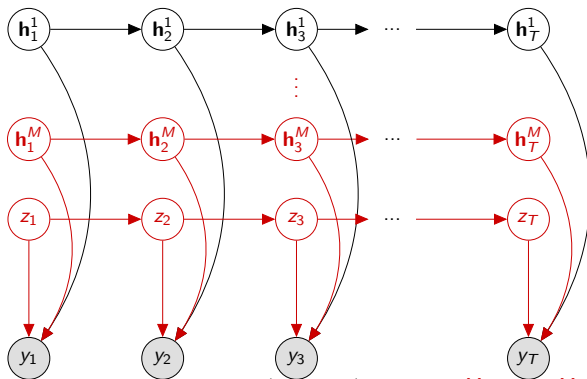
- Suppose that the time-series data can behave quite differently in different periods (**regimes**)
 - E.g. do traffic conditions evolve the same way over time in congestion periods as they do in free flow?
- How can we change the model to incorporate this prior knowledge?

Extensions: Switching linear dynamical systems (SLDSs)



- Multiple latent state chains: $\{h_1^1, \dots, h_T^1\} \dots \{h_1^M, \dots, h_T^M\}$
- At each time t , a discrete latent variable z_t selects one of the latent state chains to be "active" and produce an observation y_t

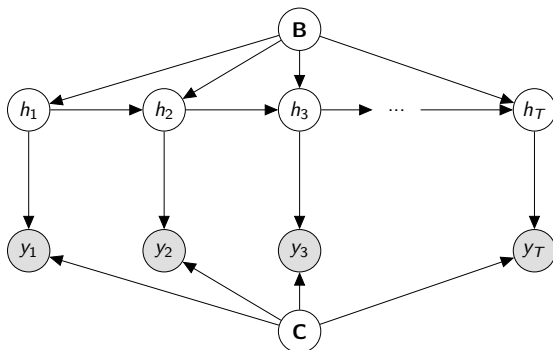
Extensions: Switching linear dynamical systems (SLDSs)



- Multiple latent state chains: $\{h_1^1, \dots, h_T^1\} \dots \{h_1^M, \dots, h_T^M\}$
- At each time t , a discrete latent variable z_t selects one of the latent state chains to be "active" and produce an observation y_t
- Useful if your observations can alternate between regimes with different dynamics
 - E.g. road segments with multiple traffic regimes (congested vs. free flow)

Hidden Markov models (HMMs)

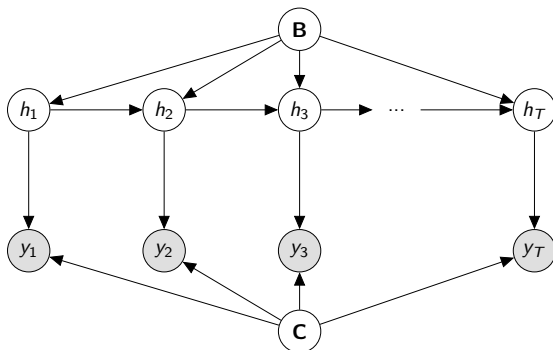
- Discrete analog of the LDS is the *hidden Markov model* (HMM)



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- Hidden states h_t and observations y_t are **discrete random variables**
- Transition probabilities $p(h_t|h_{t-1}, \mathbf{B})$ and emission probabilities $p(y_t|h_t, \mathbf{C})$ can be simply represented using conditional probability tables (CPTs)
 - Parameter matrices **B** and **C** are CPTs!

Hidden Markov models (HMMs)

- Suppose that the latent state h_t can take K possible values: $h_t \in \{1, \dots, K\}$
- We can specify the transition probabilities using a $K \times K$ matrix $\mathbf{B}$
 - Element b_{ij} of matrix $\mathbf{B}$ denotes the probability of transition to state j given that the current state is i
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- Suppose that the observations y_t can take V possible values:
 $y_t \in \{1, \dots, V\}$
- We can specify the emission probabilities using a $K \times V$ matrix $\mathbf{C}$
 - Element c_{ij} of matrix $\mathbf{C}$ denotes the probability of observing the value $y_t = j$ given that the current state h_t is i
 - Given that the current state h_t is i , the row $\mathbf{c}_i$ specifies the probabilities of observing each of the possible V values at time t ($y_t \in \{1, \dots, V\}$)
- Notice that the rows $\mathbf{b}_i$ and $\mathbf{c}_i$ must sum to 1 in order to be valid probabilities!

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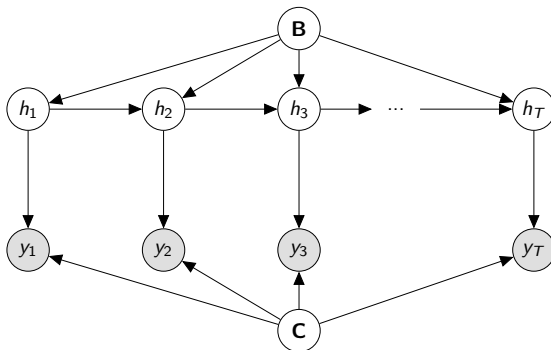
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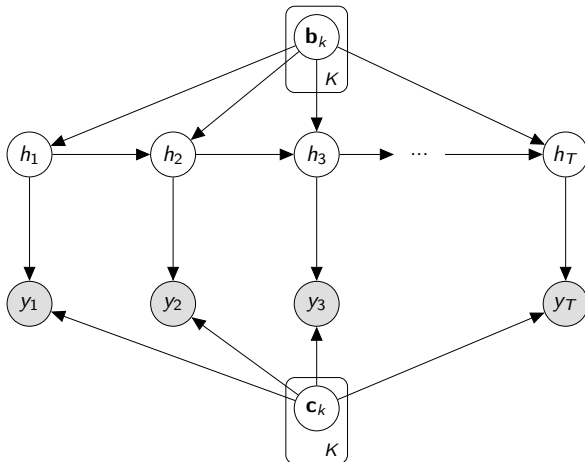
Hidden Markov models (HMMs)



- Putting everything together, the **generative process** becomes:
 - For each possible hidden state value $k \in \{1, \dots, K\}$
 - Draw transition probabilities $\mathbf{b}_k \sim \text{Dirichlet}(\mathbf{b}_k | \boldsymbol{\alpha})$
 - Draw emission probabilities $\mathbf{c}_k \sim \text{Dirichlet}(\mathbf{c}_k | \boldsymbol{\gamma})$
 - For each time $t \in \{1, \dots, T\}$
 - Draw new hidden state $h_t \sim \text{Categorical}(h_t | \mathbf{b}_{h_{t-1}})$
 - Draw observation $y_t \sim \text{Categorical}(y_t | \mathbf{c}_{h_t})$

Hidden Markov models (HMMs)

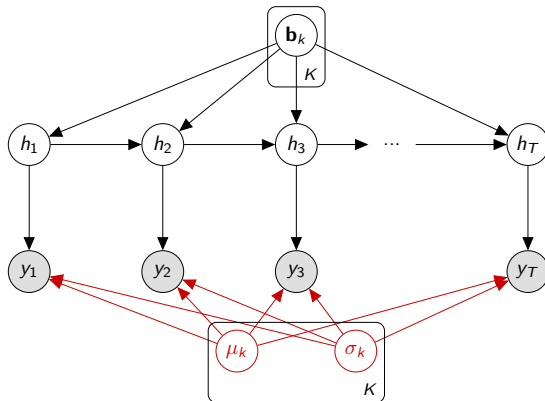
- Alternatively, we can represent the graphical model of the HMM as:



- Both ways are correct... This is one is more similar to the generative process

Extensions: HMMs with continuous observations

- We can **mix discrete** hidden states with **continuous** observations



- Hidden state value h_t defines from which of K Gaussians the observation y_t comes from

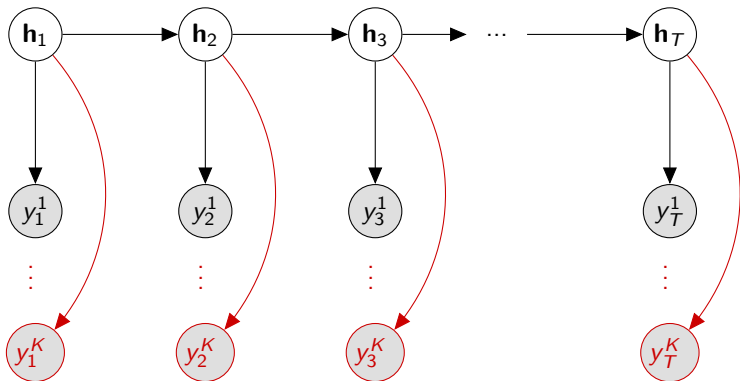
$$y_t \sim \mathcal{N}(y_t | \mu_{h_t}, \sigma_{h_t})$$

- Popular for modeling speech: y_t are sounds (continuous) and h_t are words

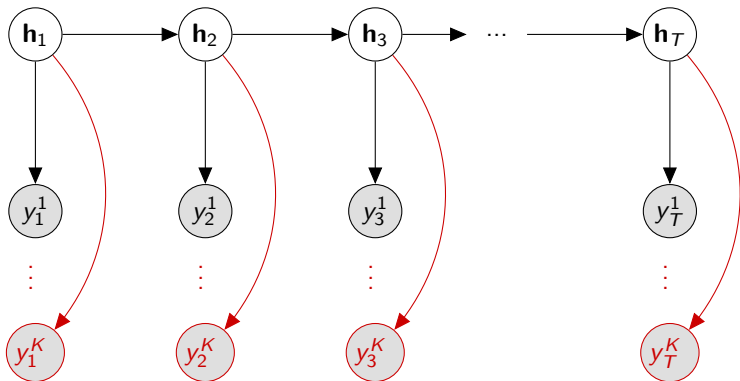
Multivariate state-space models

- What if, rather than a single time-series, we observed multiple time-series? (e.g. multiple sensors)
- Typically, multivariate time-series can exhibit **strong correlations** between the different component series
 - E.g., in our case study, occupancy rates at nearby segments of the freeway should be highly correlated
- Multivariate state-space models allow us to model these correlations

Multivariate state-space models



Multivariate state-space models



- We can group all the K observed outputs at time t in a single vector $\mathbf{y}_t = (y_t^1, \dots, y_t^K)$
- Model becomes:

$$\mathbf{h}_t \sim \mathcal{N}(\mathbf{h}_t | \mathbf{B}\mathbf{h}_{t-1}, \mathbf{R})$$

$$\mathbf{y}_t \sim \mathcal{N}(\mathbf{y}_t | \mathbf{C}\mathbf{h}_t, \mathbf{\Sigma})$$

Multivariate state-space models

- A particularly important aspect is how we define the covariance matrices $\mathbf{R}$ and $\mathbf{\Sigma}$
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- A full covariance matrix $\mathbf{\Sigma}$ also models covariance between the different elements of $\mathbf{y}$ ($N \times N$ parameters! But there are ways to reduce this...)

Multivariate state-space models

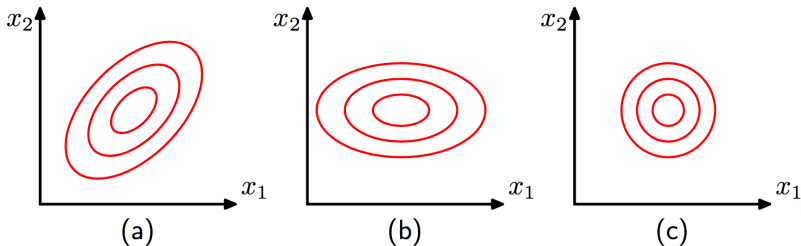


Figure: Bishop, C., 2006. Pattern Recognition and Machine Learning.

- (a) A full covariance matrix Σ (models covariances!)
- (b) A diagonal covariance of the form $\Sigma = \text{diag}(\sigma)$
- (c) An isotropic covariance of the form $\Sigma = \sigma I$

- **Multivariate** LDS model of freeway occupancy rates: multivariate forecasting
- See "06 - Temporal models - Part 3.ipynb" notebook